

Preventing Patient Falls: Addressing Fall Risks in Hospital Settings Through a Digital Patient Fall-Risk Monitoring System

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Introduction

Patient falls are a common and serious concern in hospitals because they can cause injuries, complications, longer hospital stays, and delayed recovery. Hospitalized patients may be at greater risk of falling due to factors such as old age, weakness, dizziness, poor balance, confusion, medication side effects, and limited mobility. Nurses play an important role in preventing falls by assessing patients and monitoring their conditions, but caring for multiple patients at the same time can make continuous monitoring difficult. Because of this, there is a need for an approach that can support nurses in identifying patients who are at higher risk of falling, as well as how to prevent it.

Problem Description

Patient falls are a serious patient-safety concern in hospitals because they can cause injuries, longer hospital stays, increased healthcare costs, and delayed recovery. The World Health Organization (WHO, 2024) reports that patient falls occur at an estimated 3–5 falls per 1,000 hospital bed-days, with more than one-third resulting in injury. Patients may be more likely to fall due to factors such as weakness, poor mobility, visual impairment, previous falls, and certain medications. A 2024 study found that medication use, limb weakness, unsteady walking, visual impairment, and a history of falls were important risk factors among hospitalized older adults (Yang et al., 2024). These risks can be difficult to monitor continuously, especially when nurses are caring for multiple patients.

Proposal Solution

The proposed solution is a Digital Patient Fall-Risk Monitoring System designed to assist nurses in identifying and monitoring patients who are at high risk of falling. The system can record important patient information such as age, mobility, previous falls, level of consciousness, and other relevant risk factors. It can provide alerts and

reminders to nurses when a patient requires closer observation or assistance, allowing them to respond more quickly to potential fall risks. The system can also help nurses keep track of patients who need assistance when moving, walking, or getting out of bed. By organizing fall-risk information in one accessible system, nurses can have a clearer view of which patients require priority attention.

Conclusion

Patient falls are a serious concern because they can negatively affect a patient's recovery and overall well-being. By using a digital fall-risk monitoring system, nurses can receive additional support in identifying high-risk patients and responding to potential risks more quickly. The proposed system can contribute to safer hospital environments while supporting nurses in providing effective and timely patient care.

References

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